

A Practical Approach to Proteomics

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Since the completion of the human genome project, sequencing technologies have continued to evolve, providing tools for the rapid sequencing of most model organism genomes. Associated genomic and transcriptomic data from microarray and real-time PCR technologies have yielded a wealth of new information and deeper understanding of biological systems. This genomic information has opened up the field of proteomics, allowing the identification and comparison of differentially expressed proteins, from bacteria to humans. The accumulated data show that changes in mRNA levels account for less than half of the relative expression differences observed between associated proteins, thus emphasizing the importance of proteomic data in achieving the goals of systems biology. However, with an ever-growing number of reagents, instruments, and novel technologies for the isolation, separation, and identification of proteins in complex mixtures, the task of designing appropriate proteomics experiments can be difficult. This paper describes a simple approach to unlocking the proteome of most organisms. To ensure quality data, it uses a stepwise process that combines traditional and novel reagents and instruments.

Introduction

The term proteomics was first used in 1995 and was defined as the classification of all proteins in a cell, tissue, or organism (Wilkins et al. 1996). Proteomics has since become a catchall term for virtually any research that involves proteins. For the purposes of this paper, the proteome of any cell represents all the proteins expressed at a given time. The mapping of the human genome (Lander et al. 2001) and those of other organisms has provided the primary sequence information required to assess the proteomes of biological systems. However, if splice variants and posttranslational modifications are included, the number of expressed proteins increases several times over the number of identified genes. The proteome will therefore vary in different cells and tissue types of the same organism and in different growth and developmental stages. It is also dependent on environmental factors, disease, drugs, stress, and growth conditions. Even small changes in conditions, including experimental conditions, can have significant effects on the expression, folding, and activity of proteins.

Considering all the possibilities, it is likely that any genome can potentially give rise to an infinite number of proteomes. Because proteins, not genes, are ultimately responsible for the phenotypic changes in cells and tissues, the mechanisms of disease, aging, and environmental effects cannot be elucidated solely by studying the genome. The targets of drugs and chemicals are proteins, and only through a survey of the proteome can the associated mechanisms be understood. Most importantly, the differential expression of mRNA (up or down) can capture at most 40% of the variation of protein expression (Tian et al. 2004).

The initial goal of most proteomics projects is to identify and determine differential protein expression between samples. Once a list of differentially expressed proteins has been established, the subsequent step is to perform a detailed analysis of individual proteins. This requires their expression and purification for structural characterization, assessment of biochemical activity, identification of interacting partners, or production of antibodies to quantitate expression changes. Because these analyses are time consuming and costly, accurate identification of differentially expressed proteins is critical for ensuring successful downstream analyses of individual proteins.

A typical proteomics experiment (such as protein expression profiling) can be broken down into a series of steps. First, the experiment is designed so that the key parameters of the study have been vetted, transcribed, and reviewed. Second, extraction, fractionation, and solubilization of proteins from a cell line, tissue, or organism is carried out. Third, reduce the levels of high-abundance proteins and enrich weakly expressed proteins to reduce the dynamic range in protein homogenates and increase the number of identified proteins. In the fourth step, gel-based separation of proteins in mixtures is followed by imaging and analysis to allow isolation and relative quantitation of proteins. Then gel extraction of protein spots is followed by identification by mass spectrometry; and finally, functional characterization of identified proteins is done.

These steps form the proteomics pipeline for which a rapidly growing number of reagents and instrument technologies are available for experimental use. This paper describes a simple approach to discovering differentially expressed, low-abundance proteins using a stepwise approach with validated reagents and traditional and novel technologies. This approach can provide a solid foundation for development of a small or large research program.

Materials and Methods

Protein Sample Preparation and Separation

The ProteoMiner™ protein enrichment kit (Bio-Rad Laboratories, Inc.) was used for depleting high-abundance and enriching low-abundance proteins. Spin column storage solution was removed by centrifugation, and the column beads were washed with deionized water followed by phosphate buffered saline (PBS). Human serum samples (1 ml, BioReclamation, Inc.) or *E. coli* lyophilized lysate (Bio-Rad bulletin 5656) solubilized in 1.1 ml of PBS (50 mg/ml) were applied to ProteoMiner columns, and to ensure effective binding, the columns were slowly rotated for 2 hr prior to washing away unbound proteins with PBS buffer. To elute bound low-abundance proteins, the ProteoMiner beads were treated 1–3 times with 100 µl of an acidic urea/CHAPS buffer (5% acetic acid, 8 M urea, 2% CHAPS). Then the eluted protein mixtures were treated with the ReadyPrep™ 2-D cleanup kit (Bio-Rad). Protein quantitation was performed using the Quick Start™ Bradford protein assay (Bio-Rad).

One- and Two-Dimensional Electrophoresis and Image Analysis

SDS-PAGE was performed on Criterion™ 4–20% Tris-HCl gels (Bio-Rad). Human serum and *E. coli* proteins (30 µg) from the fractions enriched by ProteoMiner technology were loaded onto the gel, separated for 1 hr at 200 V, and stained with Bio-Safe™ Coomassie stain (Bio-Rad).

For 2-D gel experiments, protein (100 µg or 200 µg) was loaded onto an 11 cm ReadyStrip™ IPG strip (Bio-Rad), pH 5–8. Isoelectric focusing (IEF) was performed using a PROTEAN® (Bio-Rad) IEF cell at 250 V for 30 min followed by 8,000 V until 45,000 V-hr were reached. The second-dimension electrophoresis was performed on a Criterion 8–16% Tris-HCl gel for 1 hr at 200 V prior to staining with Flamingo™ fluorescent gel stain (100 µg protein load) and Bio-Safe Coomassie stain (200 µg protein load).

Flamingo- and Coomassie-stained gels were imaged using the Molecular Imager® PharosFX™ and GS-800™ systems, respectively, and analyzed with PDQuest™ 2-D analysis software, version 8.0 (all from Bio-Rad).

Purification of Recombinant Proteins Under Native Conditions

All Profinity eXact™ fusion-tagged proteins used in this study were overproduced in *E. coli* and purified with Profinity eXact purification resin using either Profinity eXact mini spin columns or Bio-Scale™ Mini Profinity eXact cartridges, following protocols provided in the Profinity eXact system manual (Bio-Rad).

Results and Discussion

Given the highly dynamic nature of any proteome, a standardized approach to each experimental step is critical for reproducible and quantitative results. The quality of data produced from a proteomics experiment is directly impacted by the care and rigor employed in sample preparation, which involves the following:

Step 1: Experimental Design

Since protein expression in a cell is highly dependent on environmental alterations, proteomics experiments must be designed to ensure that all samples are treated identically. Factors that can have a major influence on the proteome include incubation time and temperature and the parameters for processing samples, such as the amount of time between tissue excision and subsequent freezing or the conditions and timing for thawing samples. Taking time to plan the experiment on paper, including a collegial review of the final design, will save months of downstream effort in troubleshooting a poorly planned experiment. This is particularly important when working with proteins, because of their dynamic nature. Consequently, a good design should detail every step in sample handling to ensure reproducible high-quality data.

Step 2: Protein Extraction

Most protocols include the following: detergents to solubilize hydrophobic membrane proteins, reductants of inter- and intraprotein disulfide bonds, denaturing agents to unfold proteins, enzymes to digest contaminating molecules (such as nucleic acids), and protease inhibitors to prevent digestion of solubilized proteins. Protein extraction may be preceded by subcellular fractionation to enrich proteins of interest localized within the cell. For example, a fractionation approach may be most appropriate for studying the proteome of the early secretory pathway, which would require enriching the endoplasmic reticulum (ER) and Golgi apparatus fractions. Bio-Rad offers a number of protein extraction and fractionation kits that perform virtually any type of fractionation for enriching the proteins of interest (Bio-Rad bulletin 3145).

Step 3: Protein Separation to Quantitate Low-Abundance Proteins

Complex protein mixtures such as serum and cell or tissue lysates contain a small number of highly abundant proteins that may mask low-abundance proteins and cause streaking on 2-D gels, which will reduce the number of proteins detected. In most proteomics experiments, the most interesting proteins are low in abundance, and a key goal is therefore to ensure that samples are treated to maximize the detection of the least concentrated proteins, since they will provide the most meaningful data.

Immunodepletion, which utilizes antibodies against high-abundance proteins, has been successfully used to remove the most abundant 6, 12, or 20 serum and plasma proteins (Echan et al. 2005, Huang et al. 2005, Zolotarjova et al. 2005). This approach involves binding selected antibodies to a chromatographic support. When serum or plasma proteins are in contact with the antibody-decorated beads, the high-abundance proteins are retained and the low-abundance proteins are eluted for use in downstream

analysis. Although this approach works quite well, its disadvantages are the high cost of antibodies, dilution of the sample, and the loss of low-abundance proteins complexed to the high-abundance proteins removed from the sample.

The novel ProteoMiner protein enrichment technology uses a combinatorial library of hexapeptides bound to a chromatographic support. It offers an alternative approach that should overcome most, if not all, of the disadvantages of immunodepletion while still effectively depleting high-abundance proteins as illustrated in Figure 1. This simple one-step technology dramatically increases the number of detected proteins and easily allows the relative quantitation of proteins in samples (Figure 2).

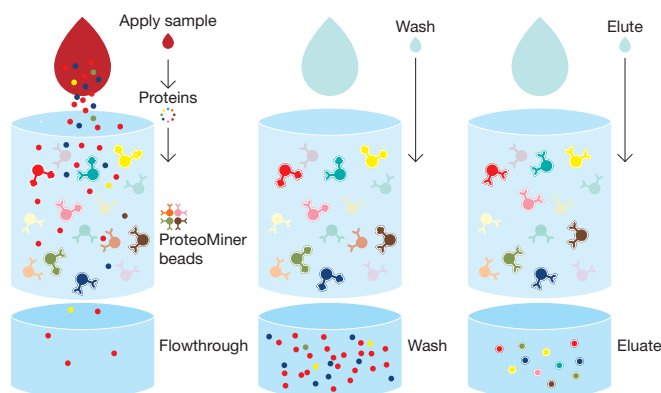


Fig. 1. Enrichment in low abundance proteins using the ProteoMiner technology. ProteoMiner technology is based on the interaction of complex protein samples with a large, highly diverse library of hexapeptides bound to chromatographic supports. Each unique hexapeptide binds to a unique protein sequence. Since the bead capacity is finite, high-abundance proteins quickly saturate their hexapeptide ligands and excess protein is washed out during the procedure. In contrast, low-abundance proteins are concentrated on their specific ligands, thereby decreasing the dynamic range of protein concentrations in the sample.

Step 4: Gel-Based Separation of Protein Mixtures, Imaging, and Analysis

Two-dimensional gel electrophoresis is a classical process commonly used today for proteomics, because its high resolving power permits the visualization of thousands of protein forms on one gel. Bio-Rad provides precast gels and buffers for both 1-D and 2-D gel electrophoresis.

It is important to consider the type of stain used for protein detection. The stain will impact both the limit of detection and the dynamic range of quantitation. Most mass spectrometers can detect proteins in the low nanogram levels; therefore, ideally the stain used will permit detection to at least this level with a broad linear range of detection. Coomassie stains commonly used for protein gels have varying degrees of sensitivity and a dynamic range limited to about one order of magnitude. Bio-Safe Coomassie stain permits detection of proteins to a limit of about 10 nanograms, with greatly reduced

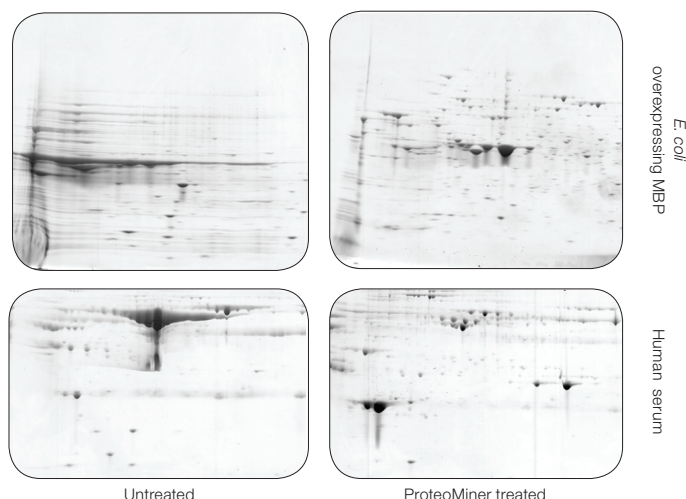


Fig. 2. ProteoMiner treatment of *E. coli* and human serum protein samples. 2-D gel analysis of *E. coli* (top panels) and human serum protein samples (bottom panels). Comparison of treated vs untreated samples shows more protein spots visible for treated samples. Gels were stained with Bio-Safe Coomassie reagent.

toxicity and consistent destaining, which produces high contrast bands in destained gels. Fluorescent stains have limits of detection in the high picogram range. The most commonly used stains include SYPRO Ruby stain and the more sensitive Flamingo stain. These stains have the additional benefit of a linear dynamic range of two to three orders of magnitude, allowing improved quantitation of proteins between gels. However, since fluorescent stains are excited by UV radiation, spot cutting instruments such as the EXQuest™ spot cutter (Bio-Rad) or an efficient UV transilluminator are required to cut protein spots from the gels.

The imaging technology used to view and quantitate protein spots on gels is also important. Many instruments with specifications to suit the needs of any research project are available to image protein gels. For 2-D protein gels, an imager with high sensitivity and high resolution is ideal. Typically these are scanning-based technologies provided by instruments such as the GS-800 densitometer or the PharosFX system, which are well suited for imaging Coomassie- or fluorescent-stained protein gels, respectively. CCD camera-based imaging technologies are less expensive and will also provide good images of 2-D gels, but with lower resolution than scanners. The Molecular Imager ChemiDoc™ XRS system (Bio-Rad) provides a good alternative to scanning technologies at lower cost.

Another important choice concerns the software used to analyze imaging data. Bio-Rad's PDQuest software is wizard-driven to automate the process of comparing the intensity of protein spots in 2-D gels. It uses a variety of statistical algorithms and user-defined preferences to analyze 2-D gel data. The approach provides a highly reproducible process for analyzing gels from multiple experiments using the same methods and settings.

The time and expense invested in sample preparation, separation, and 2-D gel electrophoresis can be significant, even for small projects. Therefore, the choice of the appropriate stain, imaging technology, and analysis software will ensure high-quality data while minimizing expense (Figure 3).

Step 5: Gel Extraction of Protein Spots for Identification by Mass Spectrometry

Differentially expressed proteins identified by PDQuest software can be marked on a printed image of the protein gel (Figure 2), and the spots can then be manually cut from the gel or automatically excised using the EXQuest spot cutter, which integrates seamlessly with PDQuest software. For manual spot picking, the OneTouch Plus spot picker (Gel Company) with disposable tips removes 1.5 mm diameter spots from 2-D gels.

There are two important factors to consider when handling 2-D gels for imaging and spot picking.

First, gels must be handled with gloves and placed on clean surfaces free of contamination. Keratin contamination from skin and hair is the leading cause of inconclusive mass spectrometry results. Use of hair nets, gloves, and masks is common practice when handling protein gels.

Second, cut the gels in a clean, HEPA-filtered fume hood or via an automated spot cutter with a contained environment. Additional caution must be taken when manually cutting spots from fluorescent-stained 2-D gels using a UV transilluminator to ensure that skin and eyes are protected during the excision process, which can take anywhere from a few minutes to several hours, depending on the number of spots excised.

Step 6: Functional Characterization of Identified Proteins Through Protein Expression, Biochemical Analysis, and Antibody Production

For any proteomics project, the research begins where the project ends with a list of identified, differentially expressed proteins. ExPASy Proteomics tools (<http://ca.expasy.org/tools/>) is an amalgamation of the most popular Web-based tools for in silico characterization of proteins based on their primary sequences. Taking time to filter a list of proteins to find those that make sense within the study parameters is worthwhile, because the next steps involve the biochemical characterizations of the short-listed proteins, which can be time consuming and costly. The top candidate proteins or protein domains can then be expressed for subsequent biochemical and structural analysis and potentially for antibody production.

Other Considerations

Host cell system: For large proteins (>100 kD), a eukaryotic system is recommended, whereas small proteins (<30 kD) can be easily expressed in prokaryotic systems. If glycosylation or other posttranslational modifications are important, yeast, baculovirus, or other eukaryotic systems are preferred. With isotopic labeling, expression in *E. coli* is required. There are many organizations supplying a variety of host cell systems, including American Type Culture Collection (<http://www.atcc.org>); Clontech cell lines (<http://www.clontech.com>); Stratagene cells (*E. coli* BL21) (<http://stratagene.com>); and Invitrogen cell lines (*Pichia pastoris*) (<http://www.invitrogen.com>).

Expression vector: The choice of a vector will be dictated by the selected host cell system. For eukaryotic and prokaryotic expression, there are many expression vectors with different promoters including arabinose systems (pBAD), phage T7 (pET), Trc/Tac promoters, and phage lambda PL or PR. Proteins can be expressed fused with an affinity tag to assist in their downstream purification. The most common tags are His6 for metal affinity chromatography, FLAG epitope tag DYKDDDDK, CBP-calmodulin

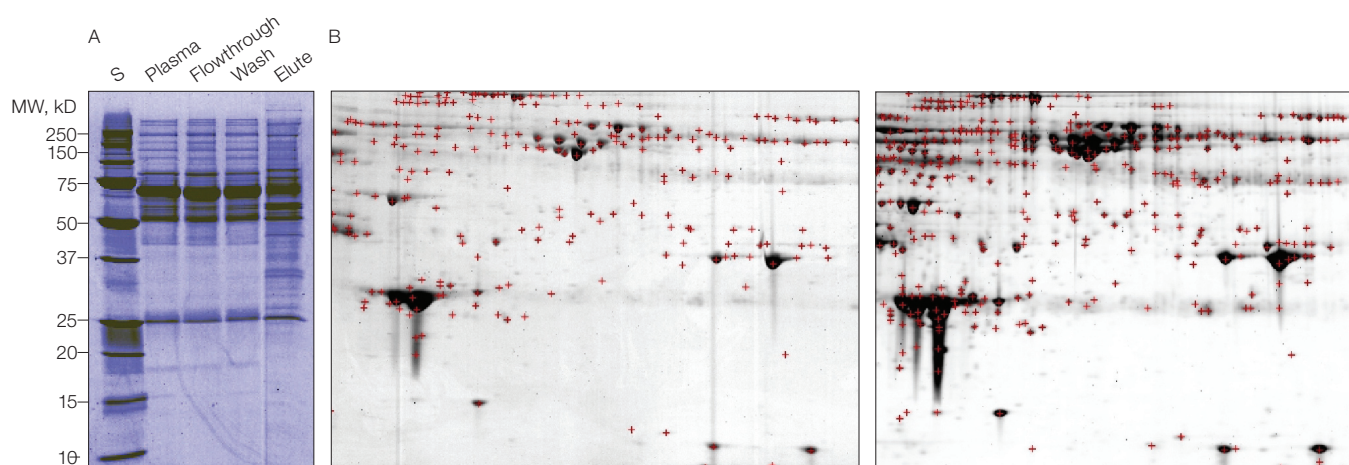


Fig. 3. Enrichment in low-abundance proteins using the ProteoMiner protein enrichment kit.

A, Coomassie-stained 1-D gel of ProteoMiner kit-treated serum sample shows an increased number of visible bands in the eluate fraction. B, Comparison of Coomassie (left panel) versus Flamingo (right panel) stains on ProteoMiner kit-treated serum samples on 2-D gel. A human serum sample was enriched in low-abundance proteins using the ProteoMiner protein enrichment kit. Spot identification was performed using PDQuest software with identical spot detection parameters. There were 249 and 402 spots detected in the Coomassie- and Flamingo-stained gels, respectively.

binding peptide (26 residues), E-coil/K-coil tags (poly E35 or poly K35), c-myc epitope tag EQKLISEEDL, glutathione-S-transferase tags, and cellulose binding domain tags. The major problem with all of these affinity tags is that they result in a modified protein that can have different biochemical or structural properties than those of the native protein, making them of limited use for in vitro characterization (Araújo et al. 2000, Amau et al. 2006, Chant et al. 2005). Furthermore, these tags are difficult or impossible to cleave from the protein without concomitant cleavage of the protein itself. Bio-Rad offers a novel tag, Profinity eXact tag, that addresses both of these issues.

The Profinity eXact fusion tag system offers a one-step purification/cleavage protocol for bacterial recombinant protein production. The system is comprised of Profinity eXact purification resin and the Profinity eXact tag, a small 8 kD polypeptide expressed as an N-terminal fusion to the target protein. The ligand coupled to the resin matrix is based on the bacterial protease subtilisin BPN', which has been extensively engineered to increase its stability and to separate the substrate binding and proteolytic functions of the enzyme (Abdulaev et al. 2005, Ruan et al. 2004). These modifications allow for highly selective binding to the chromatographic medium and specific and controlled cleavage of the tag. Cleavage is triggered by the addition of low concentrations of small anions such as fluoride or azide. The native recombinant protein is released without any residual amino acids at the N-terminus, and the 8 kD Profinity eXact tag remains bound to the modified subtilisin ligand linked to the resin. Purification of fusion proteins is performed under native conditions; tag cleavage and elution of purified protein from the column is completed in ~1 hr (Figure 4).

Is it membrane or water soluble? Membrane protein genes are very challenging to clone and express, and purification of such recombinant proteins is difficult because they are very hydrophobic. Potential problems can be avoided by knowing whether the protein contains one or more membrane-spanning helices and where these helices are located. Solubility depends on many factors, including size (smaller ones are more soluble), hydrophobicity (average and local hphob), 3-D structure and ligand interactions, overall charge, predicted accessibility, and distribution and frequency of amino acids. All of these can be examined on the ExPASy Proteomics tools website. A potential approach for characterizing a protein with hydrophobic domains is to only express and purify the soluble fragments for subsequent biochemical and structural analysis.

Is it single-domain or multidomain? Many eukaryotic proteins are multidomain; their size is a good indicator, with roughly one domain for every 15 kDa. Small domains generally behave better for both X-ray crystallography and NMR structural analysis.

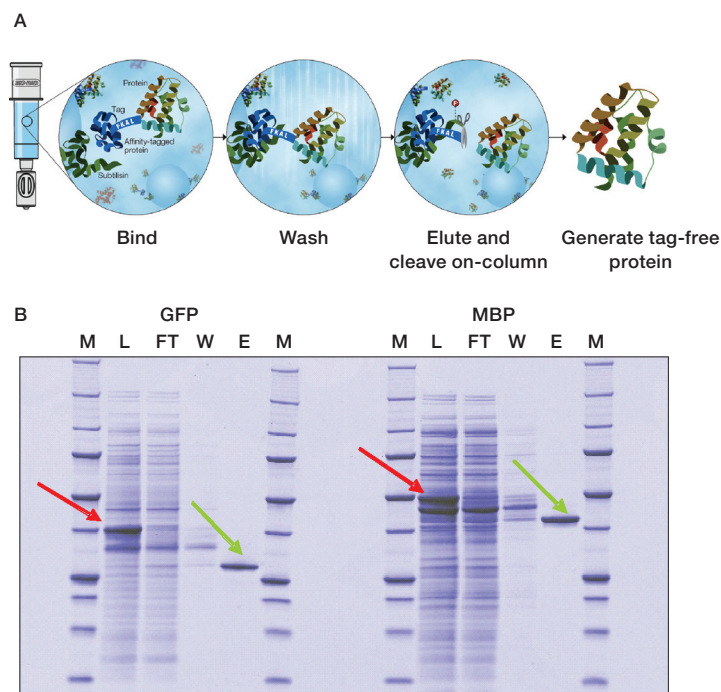


Fig. 4. Profinity eXact system purification of a protein expressed in *E. coli*. SDS-PAGE analysis of purification of a 26 kD green fluorescent protein (GFP) and a 40 kD maltose binding protein (MBP). Profinity eXact-tagged gene fusions were run on a BioLogic DuoFlow™ system (Bio-Rad). Crude *E. coli* lysate (2 ml) was loaded onto a Bio-Scale Mini Profinity eXact 1 ml cartridge with bind/wash buffer (0.1 M potassium phosphate, pH 7.2) at a flow rate of 1 ml/min. The cartridge was washed with 10 column volumes (CV) of the same buffer at 1 ml/min. Proteins were eluted with 3 CV of potassium fluoride buffer at room temperature at 0.1 ml/min for 30 min. Total purification time to generate tag-free proteins, without the addition of protease, was approximately 60 min.

Limited proteolysis allows the experimental identification of domains prior to structural determination. There are websites available to perform virtual domain prediction, such as PredictProtein (<http://www.predictprotein.org>). BLAST alignments can be used to detect or predict the presence of domains by sequence homology. Protein domains can also be predicted using the Conserved Domain Database (CDD) (<http://www.ncbi.nlm.nih.gov/Structure/cdd/cdd.shtml>).

Where will this protein be found? By knowing where the protein is found in the cell, we can glean information about its function and potential interaction partners. Is it exported? Does it go to the nucleus? Does it go through the ER? Does it localize to mitochondria or to chloroplasts? Does it go to the membrane? How do you tell? There are websites (<http://psort.nibb.ac.jp>, <http://www.cbs.dtu.dk/services/TargetP/#submission>, for example) that use the primary protein sequence to identify signal or domain sequences that predict the location of the protein of interest in the cell.

Functional characterization of identified proteins: Peptide-based or whole protein-based antibodies can be raised to and used as tools for examining the interactions between proteins, using techniques such as immunoprecipitation and western blotting. Furthermore, these antibodies can be used to detect and quantitate the amount of target protein in a sample using the Bio-Plex® suspension array system (Bio-Rad) (Jenmalm et al. 2003). The Bio-Plex suspension array system is a flexible, easy-to-use bioassay system for the simultaneous detection and quantitation of up to 100 different analytes in a single microplate well. Multiplex analysis with the Bio-Plex system yields data that are linked within a system so that complex relationships and pathways of biomolecules can be revealed. The technology requires as little as 12 µl of serum or plasma and up to 50 µl of other types of biological analytes per multiplex assay to dramatically increase the amount of useful data per sample. The system is well suited for immunoassays, enzyme assays, receptor-ligand assays, DNA hybridization assays, and RNA quantitation.

The functional characterization of proteins requires multiple approaches, including protein expression, purification, and production of antibodies to proteins of interest. With these tools, many biochemical techniques can be used to further characterize proteins and their interacting partners (Figeys 2004).

Conclusion

The greatest challenge for proteomics technology is the inherently complex nature of cellular proteomes. Different cells within a multicellular organism have different proteomes, and the number of proteins in a proteome is very large. Each proteome contains proteins that are structurally diverse and with various physicochemical characteristics. In addition, in protein interaction studies, native conformations of proteins must be maintained to obtain meaningful results. Because of these considerations, comprehensive characterization of cellular proteomes is a complicated undertaking and must be performed in a rigorous, stepwise process, as described in this report.

Although no precise calculations can be made, it is estimated that up to 50,000 protein species may be simultaneously present in a eukaryotic cell (Hanash 2000). The dynamic range of protein expression spans seven or eight orders of magnitude. Consequently, proteins are present in vastly different quantities, and many important classes of proteins (which may be important drug targets) such as transcription factors, protein kinases, and regulatory proteins are low-abundance proteins. These low-abundance proteins will not be observed in the analysis of crude cell lysates without a purification step, such as that afforded by ProteoMiner technology.

Despite the technical difficulties, proteomics, when combined with other complementary studies such as genomics, has enormous potential to provide new biological insights. The ability to study complex biological systems in their entirety will ultimately provide answers that cannot be obtained from the study of individual proteins or groups of proteins. Combined with gene expression data, proteomics will allow biologists to build a more complete model of the systems they study, and with continued advances in technology, the discovery process will advance rapidly.

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Expression and purification of GST fusion proteins may require a license under US patent 5,654,176 (assignee: Chemicon International).

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